

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1.

You provide access to your model's source code and document the data sources you used for training an AI model. This is an example of:

1 / 1 point
- ☐ Explainability

☒ Transparency

☐ Interpretability

🕒 Correct

Correct! Transparency involves providing documentation for the system's decisions and actions (details about model architecture, training data, optimizations, etc.) to ensure compliance and accountability

2.

Why is explainability important for AI systems? (Check all that apply):

1 / 1 point

☒ Trust and Accountability

🕒 Correct

Correct! Being able to understand and audit an AI's decision-making process is crucial for building trust, especially in high-stakes domains like healthcare, criminal justice, and finance.

☒ Identifying Biases

🕒 Correct

Correct! Explainable models can help uncover unintended biases, discrimination, or other ethical issues embedded in the training data or model.

☒ Human-AI Collaboration

🕒 Correct

Correct! When humans can understand an AI's reasoning, they can provide guidance, feedback and corrections to improve future outputs. Debugging AI systems is part of this collaboration and explainable methods make debugging easier.

☒ Scientific Advancement

🕒 Correct

Correct! Explainable models facilitate deeper understanding of the underlying phenomena being modeled.

3.

Which of the following are NOT strategies for Responsible AI?

1 / 1 point

☐ Transparency and Explainability

☐ Robustness and Safety

☐ Fairness and Non-discrimination

☒ Monetization and Technical Superiority

🕒 Correct

Correct! While Responsible AI is not explicitly against monetization or technical superiority, these are not strategies of Responsible AI.

4.

An image recognition system designed to identify faces is trained primarily on images of people with lighter skin tones. As a result, the system performs poorly when trying to recognize faces with darker skin tones. What type of algorithmic bias is this?

1 / 1 point

☒ Training Data Bias

☐ Sample Bias

☐ Bias resulting from the algorithms themselves

🕒 Correct

Correct! If the data used to train an AI model is biased or lacks representation from certain groups, the model may learn and perpetuate those biases.

5.

An algorithm designed to predict job performance uses the zip code of applicants as one of its features. What type of algorithmic bias could this lead to?

1 / 1 point

☐ Bias arising from human-AI interaction

☒ Proxy Variables Bias

☐ Training Data Bias

🕒 Correct

Correct! Zip codes can act as a proxy for race or socioeconomic status, leading to biased predictions that disproportionately affect candidates from disadvantaged areas, reinforcing existing inequalities.

6.

A healthcare predictive model is trained on data from a specific hospital that primarily serves a younger population. When this model is used in a different hospital that treats a more varied age group, its predictions are less accurate for elderly patients. What type of algorithmic bias is this?

1 / 1 point

☒ Sample Bias

☐ Training Data Bias

☐ Bias resulting from the algorithms themselves

🕒 Correct

Correct! Because the sample from the first hospital was not representative of the broader population, the model was not generalizable to the other hospital with different age demographics.

7.

What is AI Alignment?

1 / 1 point

☐ Ensure advanced AI systems remain aligned with evaluation criteria set by human-AI collaboration

☐ Ensure advanced AI systems remain aligned with other advanced AI systems

☒ Ensure advanced AI systems remain aligned with human values, intentions and ethics

🕒 Correct

Correct! AI Alignment refers to the challenge of ensuring advanced AI systems remain aligned with human values, intentions and ethics as they become more capable and autonomous.

8.

Approaches like LIME and SHAP are examples of:

1 / 1 point

☐ Transparency

☒ Explainability

☐ Interpretability

☐ Alignment

🕒 Correct

Correct! Explainable methods include Local explainable techniques like SHAP and LIME. We will learn more about them in the next module.

9.

When we say black box models, do we mean they are not reproducible?

1 / 1 point

☐ Yes

☐ No

☒ Sometimes

🕒 Correct

Correct! Not all black box models are reproducible, but a black box model does not mean a model in which we cannot reproduce the result. If we have the model architecture including trained weights, the same input data, and the random seed used to produce the output (for models containing some stochastic process, such as dropout layers or certain types of attention mechanisms), then we can reproduce the result. The idea around black box models is that while we may have the trained weights, these weights are not understandable by humans.

10.

In her TED talk, Dr. Buolamwini shares a striking example that shows algorithmic bias and what she terms the "coded gaze". What is this example?

1 / 1 point

☒ The facial detection system is unable to detect Dr. Buolamwini's face until she puts on a white Halloween mask.

☐ The facial detection system is unable to detect Dr. Buolamwini's face when the lights are off.

☐ The facial detection system is unable to detect Dr. Buolamwini's face until she removes her glasses.

🕒 Correct

Correct! This shows a striking example of algorithmic bias in facial recognition systems. Although this talk was recorded in 2016, many of these facial recognition technologies are still being used in fields like criminal justice.